MATH1231 Mathematics 1B – Algebra

Section 04 - Linear Combinations and Spans

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Linear combinations

Definition (Linear combination)

A **linear combination** of vectors $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$ in a vector space V is any expression of the form

$$\lambda_1 \overrightarrow{\mathbf{v_1}} + \lambda_2 \overrightarrow{\mathbf{v_2}} + \cdots + \lambda_n \overrightarrow{\mathbf{v_n}} ,$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are scalars.

Note that if $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$ are in a vector space V, then every linear combination of these vectors is also in V. (Why?)

Examples 1.

a) We have $\begin{pmatrix} -1 \\ -3 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 1 \end{pmatrix} - 3\begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$
so $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$ is a linear combination of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$.

b) Also, $7 - 2x + 2x^2 = 3(-1 + x + 4x^2) + 5(2 - x - 2x^2)$,
so $7 - 2x + 2x^2$ is a linear combination of the polynomials $-1 + x + 4x^2$ and $2 - x - 2x^2$.

Linear combinations, continued

Example 2. Is $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ a linear combination of $\mathbf{v_1} = \begin{pmatrix} 9 \\ -1 \\ 6 \end{pmatrix}$, $\mathbf{v_2} = \begin{pmatrix} \sqrt{2} \\ 3 \\ -\sqrt{3} \end{pmatrix}$ and $\mathbf{v_3} = \begin{pmatrix} -\sqrt{3} \\ \sqrt{2} \\ 5 \end{pmatrix}$?

Linear combinations, continued

Example 3. Is $\mathbf{u} = \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}$ a linear combination of $\mathbf{u_1} = \begin{pmatrix} 2 \\ 0 \\ -5 \end{pmatrix}$, and $\mathbf{u_2} = \begin{pmatrix} 0 \\ 1 \\ 7 \end{pmatrix}$?

Linear combinations, continued

Example 4. Consider the matrices $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $A_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

a) Is the matrix $B = \begin{pmatrix} -1 & 5 \\ 5 & 0 \end{pmatrix}$ a linear combination of the matrices A_1, A_2 and A_3?

b) Is the matrix $C = \begin{pmatrix} 2 & 9 \\ 3 & 1 \end{pmatrix}$ a linear combination of the matrices A_1, A_2 and A_3?

c) Describe all linear combinations of A_1, A_2, A_3.

Span

Definition (Span)

The **span** of a set of vectors $S = \{\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}\}$ in a vector space V, written

$$\operatorname{span}(S) \quad \text{or} \quad \operatorname{span}(\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}),$$

is the set of all linear combinations of these vectors.

Note that $S \subseteq \operatorname{span}(S) \subseteq V$.

Example 5.

a. Describe geometrically the span of $S = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$.

Solution. Here $\operatorname{span}(S) = \left\{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} : \lambda \in \mathbb{R} \right\}$ is the line through the origin and the point $(1, 2)$ in $\mathbb{R}^2$.

Example 5: Examples of spans, continued.

b. Describe geometrically the span of $S = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}$.

Solution. Here $\text{span}(S) = \left\{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \end{pmatrix} : \lambda, \mu \in \mathbb{R} \right\}$ is the plane through the origin and the points $(1, 2)$ and $(2, 3)$ in $\mathbb{R}^2$ (that is, all of $\mathbb{R}^2$).

c. Describe geometrically the span of $S = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix} \right\}$.

Solution. Here
$$\begin{aligned} \text{span}(S) &= \left\{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 4 \end{pmatrix} : \lambda, \mu \in \mathbb{R} \right\} \\ &= \left\{ (\lambda + 2\mu) \begin{pmatrix} 1 \\ 2 \end{pmatrix} : \lambda, \mu \in \mathbb{R} \right\} \end{aligned}$$
is the line through the origin and the point $(1, 2)$ in $\mathbb{R}^2$.

d. Let $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $A_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. What is $\text{span}(A_1, A_2, A_3)$ in $M_{2,2}(\mathbb{R})$?

Solution. $\text{span}(A_1, A_2, A_3)$ is the set of all symmetric 2×2 matrices with real entries, because taking linear combinations of the given matrices will give *all* symmetric 2×2 matrices *and no others*.

Example 5: Examples of spans, continued.

e. Consider $S = \{1, x, x^2\}$, a subset of $\mathbb{P}(\mathbb{R})$.
(Recall that $\mathbb{P}(\mathbb{R})$is the set of all polynomials with real coefficients.)

(i) What is span(S)?

Solution. Any linear combination of S is of the form $a_0 + a_1 x + a_2 x^2$ for some $a_0, a_1, a_2 \in \mathbb{R}$. Since $\mathbb{P}_2$ is the set of all polynomials of this form, we can say span$(S) = \mathbb{P}_2$.

Note: We used $\mathbb{P}_2$ as shorthand for $\mathbb{P}_2(\mathbb{R})$ here. We shall often write just $\mathbb{P}$ instead of $\mathbb{P}(\mathbb{F})$ if it is clear from the context what scalar field we are talking about.

(ii) Is x^3 in the span of $S = \{1, x, x^2\}$?

Solution. Here $x^3 \notin$ span(S), since x^3 cannot be written as a linear combination $a_0 + a_1 x + a_2 x^2 = x^3$ for any $a_0, a_1, a_2 \in \mathbb{R}$.

Alternatively, we could say given the previous example that $x^3 \notin \mathbb{P}_2$.

The span of anything is a subspace

In the last four examples, the span of various numbers of vectors has turned out to be a vector space. In fact, this is always true.

Fact (Spans are subspaces)

Let V be a vector space; let $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$ be vectors in V, and write $W = \operatorname{span}(\overrightarrow{\mathbf{v}}_1, \overrightarrow{\mathbf{v}}_2, \dots, \overrightarrow{\mathbf{v}}_n)$. Then W is a subspace of V. In fact, W is the *smallest* subspace of V which contains $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$.

Proof. We use the subspace theorem.

◇ As we mentioned earlier, every linear combination of vectors in V is also in V; thus $\operatorname{span}(\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}})$ is a subset of V; and it is given that V is a vector space.

◇ The zero vector $\overrightarrow{\mathbf{0}}$ is in W because it is a linear combination of the given vectors, $\overrightarrow{\mathbf{0}} = 0\overrightarrow{\mathbf{v_1}} + 0\overrightarrow{\mathbf{v_2}} + \cdots + 0\overrightarrow{\mathbf{v_n}}$.

◇ Now suppose that $\overrightarrow{\mathbf{w}}$ and $\overrightarrow{\mathbf{x}}$ are in W. By the definition of span, we have

$$\overrightarrow{\mathbf{w}} = \lambda_1\overrightarrow{\mathbf{v_1}} + \lambda_2\overrightarrow{\mathbf{v_2}} + \cdots + \lambda_n\overrightarrow{\mathbf{v_n}} \qquad \overrightarrow{\mathbf{x}} = \mu_1\overrightarrow{\mathbf{v_1}} + \mu_2\overrightarrow{\mathbf{v_2}} + \cdots + \mu_n\overrightarrow{\mathbf{v_n}}$$

for some scalars $\lambda_1, \lambda_2, \dots, \lambda_n, \mu_1, \mu_2, \dots, \mu_n$. By using the associative, commutative and distributive laws in V we obtain

$$\overrightarrow{\mathbf{w}} + \overrightarrow{\mathbf{x}} = (\lambda_1 + \mu_1)\overrightarrow{\mathbf{v_1}} + (\lambda_2 + \mu_2)\overrightarrow{\mathbf{v_2}} + \cdots + (\lambda_n + \mu_n)\overrightarrow{\mathbf{v_n}} \,,$$

which is a linear combination of $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$ and is therefore in W. Thus W is closed under addition.

The span of anything is a subspace, continued

◇ Finally, let $\overrightarrow{\mathbf{w}}$ be in W and let λ be a scalar. Then

$$\overrightarrow{\mathbf{w}} = \lambda_1 \overrightarrow{\mathbf{v_1}} + \lambda_2 \overrightarrow{\mathbf{v_2}} + \cdots + \lambda_n \overrightarrow{\mathbf{v_n}}$$

for some scalars $\lambda_1, \lambda_2, \dots, \lambda_n$, and so

$$\lambda \overrightarrow{\mathbf{w}} = (\lambda\lambda_1)\overrightarrow{\mathbf{v_1}} + (\lambda\lambda_2)\overrightarrow{\mathbf{v_2}} + \cdots + (\lambda\lambda_n)\overrightarrow{\mathbf{v_n}} \ ,$$

which again is a linear combination of $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$. Therefore W is closed under scalar multiplication.

We have shown that W is a subset of the given vector space V, and that W contains the zero vector and is closed under addition and scalar multiplication. Hence, W is a subspace of V.

◇ To show that W is the *smallest* subspace of V containing $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$, let X be *any* subspace of V containing these vectors: we want to show that W is a subset of X. But this is true because every linear combination of $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$ is in X, and so every element of W is in X. This completes the proof.

Yet another way to prove something is a vector space

Besides its theoretical importance, the previous result can be used to provide a very short and efficient proof that a given set is a vector space. We re–do a previous example.

Example 6. Show that $S = \{(x_1,\ 2x_2,\ x_2,\ x_1 + x_2) \mid x_1, x_2 \in \mathbb{R}\}$ is a vector space.

Solution. We have
$$\begin{aligned} S &= \{x_1(1,0,0,1) + x_2(0,2,1,1) \mid x_1, x_2 \in \mathbb{R}\} \\ &= \operatorname{span}\{(1,0,0,1),\ (0,2,1,1)\} \end{aligned}$$
and this is a vector space because the span of *any* (finite) collection of vectors is a vector space.

Next, we want to approach the previous problem from a slightly different point of view. Given $\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}$ in a vector space V, we have been trying to find out exactly what $\text{span}\{\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}\}$ is.

Problem

Now we only want to know if $\text{span}\{\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}\}$ equal to the whole vector space V, or just a part of V.

Definition (spanning set)

For vectors $\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}$ in a vector space V, the following statements all mean the same thing:

- every vector in V is a linear combination of $\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}$;
- $\text{span}\{\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}\} = V$;
- $\{\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}\}$ is a **spanning set** for V;
- $\{\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}\}$ **spans** V.

The point is: we know that $\text{span}\{\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}\}$ is a subspace of V; this subspace may be the whole of V, or it may be only part of V. The former case is what is meant by saying "$\{\vec{\mathbf{v_1}}, \vec{\mathbf{v_2}}, \ldots, \vec{\mathbf{v_n}}\}$ spans V".

Problem Do *two* vectors span $\mathbb{R}^3$? Sometimes? Always? Never?

Never. Two vectors $\vec{\mathbf{v}}$, $\vec{\mathbf{w}}$ in $\mathbb{R}^3$ cannot span $\mathbb{R}^3$.
We saw that span($\vec{\mathbf{v}}$, $\vec{\mathbf{w}}$) may be $\{\vec{\mathbf{0}}\}$, a line through the origin, or a plane through the origin, but will never be the whole of $\mathbb{R}^3$.

Problem Do *three* vectors span $\mathbb{R}^3$? Sometimes? Always? Never?
Let's look at some examples.

Examples 7.

a) The three unit vectors $\vec{\mathbf{i}}$, $\vec{\mathbf{j}}$ and $\vec{\mathbf{k}}$ do span $\mathbb{R}^3$, because *any* vector

$$\vec{\mathbf{b}} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

in $\mathbb{R}^3$ can be written as a linear combination of $\vec{\mathbf{i}}$, $\vec{\mathbf{j}}$, $\vec{\mathbf{k}}$, namely,

$$\vec{\mathbf{b}} = b_1 \vec{\mathbf{i}} + b_2 \vec{\mathbf{j}} + b_3 \vec{\mathbf{k}} \ .$$

Spanning sets for $\mathbb{R}^3$

Problem So do three vectors *always* span $\mathbb{R}^3$?

b) Consider three vectors which are scalar multiples of each other, for example,

$$\vec{\mathbf{u}} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}, \quad \vec{\mathbf{v}} = \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}, \quad \vec{\mathbf{w}} = \begin{pmatrix} -5 \\ 10 \\ -15 \end{pmatrix}.$$

The span of these vectors is only a line, not the whole of $\mathbb{R}^3$. Therefore $\{\vec{\mathbf{u}}, \vec{\mathbf{v}}, \vec{\mathbf{w}}\}$ is not a spanning set for $\mathbb{R}^3$.

Spanning sets for $\mathbb{R}^3$

c) Harder: Do the vectors

$$\overrightarrow{\mathbf{u}} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}, \quad \overrightarrow{\mathbf{v}} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}, \quad \overrightarrow{\mathbf{w}} = \begin{pmatrix} -2 \\ -3 \\ 7 \end{pmatrix}$$

span $\mathbb{R}^3$?

Answer.We take an arbitrary vector $\overrightarrow{\mathbf{b}}$ in $\mathbb{R}^3$ and attempt to write it as a linear combination

$$\overrightarrow{\mathbf{b}} = \lambda_1 \overrightarrow{\mathbf{u}} + \lambda_2 \overrightarrow{\mathbf{v}} + \lambda_3 \overrightarrow{\mathbf{w}} \ .$$

Obviously we can find $\lambda_1, \lambda_2, \lambda_3$ in some cases (for example, when $\overrightarrow{\mathbf{b}} = \cdots$?) But the important question is, can we find $\lambda_1, \lambda_2, \lambda_3$ in *all* cases, no matter what $\overrightarrow{\mathbf{b}}$ is? Writing the vector equation in the form of three scalar equations, we have

$$\lambda_1 - 2\lambda_2 - 2\lambda_3 = b_1 \ , \quad \lambda_2 - 3\lambda_3 = b_2 \ , \quad -2\lambda_1 + 4\lambda_2 + 7\lambda_3 = b_3 \ .$$

We put these equations into augmented matrix form and row–reduce:

$$(A|\overrightarrow{\mathbf{b}}) = \left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & 1 & -3 & b_2 \\ -2 & 4 & 7 & b_3 \end{array}\right) \longleftrightarrow (U|\overrightarrow{\mathbf{y}}) = \left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & 1 & -3 & b_2 \\ 0 & 0 & 3 & 2b_1 + b_3 \end{array}\right) .$$

Spanning sets for $\mathbb{R}^3$, continued

$$(U|\mathbf{y}) = \left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & 1 & -3 & b_2 \\ 0 & 0 & 3 & 2b_1 + b_3 \end{array}\right)$$

• No matter what b_1, b_2 and b_3 are, the last column y is not a leading column, so the system has at least one solution;

• therefore, the system has a solution not only for some right hand sides, but for **every** right hand side;

• therefore, $\{\overrightarrow{\mathbf{u}}, \overrightarrow{\mathbf{v}}, \overrightarrow{\mathbf{w}}\}$ is a spanning set for $\mathbb{R}^3$.

Comment. It is strongly recommended that in this type of problem you give a three–part reason for your answer: (i) what you see in the echelon form; (ii) what this says about solutions of the system of equations; (iii) how this answers the question asked.

Spanning sets, another example.

Exercise 8. Do the vectors

$$\overrightarrow{\mathbf{u}} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \quad \overrightarrow{\mathbf{v}} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}, \quad \overrightarrow{\mathbf{w}} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$$

span $\mathbb{R}^3$? If not, find condition(s), if any, on b_1, b_2, b_3 for $\overrightarrow{\mathbf{b}}$ to be in the span of $\overrightarrow{\mathbf{u}}, \overrightarrow{\mathbf{v}}, \overrightarrow{\mathbf{w}}$. Give a detailed geometric interpretations of $\text{span}\{\overrightarrow{\mathbf{u}}, \overrightarrow{\mathbf{v}}, \overrightarrow{\mathbf{w}}\}$.

Span of sets of vectors in $\mathbb{R}^m$ and matrices

Preparation for the big result to come.

Exercise 9.

Consider the matrix $A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$ and the vector $\overrightarrow{\mathbf{x}} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$.

Fill in the blanks: $A\overrightarrow{\mathbf{x}} = x_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + x_2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} + x_3 \begin{pmatrix} 5 \\ 6 \end{pmatrix}$.

In plain English, this result says that $A\overrightarrow{\mathbf{x}}$ can be interpreted as a linear combination where

- the vectors in the linear combination are the columns of A,
- the coefficients of the linear combination are the components of $\overrightarrow{\mathbf{x}}$.

Span of sets of vectors in $\mathbb{R}^m$ and matrices

TIP! *The key here is to keep in mind that $\overrightarrow{\mathbf{b}} = A\overrightarrow{\mathbf{x}}$, means that $\overrightarrow{\mathbf{b}}$ is a linear combination of the columns of A, with scalar coefficients $x_1, x_2, \cdots, x_n$, the components of $\overrightarrow{\mathbf{x}}$.*

Fact (Matrices, linear combinations and spans in $\mathbb{R}^m$.)

Let $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$ be vectors in $\mathbb{R}^m$, and let A be the $m \times n$ matrix whose columns are $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}$. Then

- a specific vector $\overrightarrow{\mathbf{b}} \in \mathbb{R}^m$ is a linear combination of the vectors $\{\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}\}$ if and only b can be written $\overrightarrow{\mathbf{b}} = A\overrightarrow{\mathbf{x}}$ for some vector $\overrightarrow{\mathbf{x}}$ in $\mathbb{R}^n$;
- a specific vector $\overrightarrow{\mathbf{b}} \in \mathbb{R}^m$ belongs to $\text{span}(\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}})$ if and only the equation $A\overrightarrow{\mathbf{x}} = \overrightarrow{\mathbf{b}}$ has a solution in $\mathbb{R}^n$.
- Corollary: $\{\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \dots, \overrightarrow{\mathbf{v_n}}\}$ spans $\mathbb{R}^m$ if and only if $A\overrightarrow{\mathbf{x}} = \overrightarrow{\mathbf{b}}$ has a solution for *every* $\overrightarrow{\mathbf{b}}$ in $\mathbb{R}^m$.

Definition (column space of a matrix)

Let A be an $m \times n$ matrix with entries in a field $\mathbb{F}$. The span of the columns of A is called the **column space** of A and is denoted by **col(A)**.

In other words, if A has columns $\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \ldots, \overrightarrow{\mathbf{v_n}}$, then $\mathrm{col}(A)$ and $\mathrm{span}\{\overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}}, \ldots, \overrightarrow{\mathbf{v_n}}\}$ are just two names for the same thing.

Note that since $\mathrm{col}(A)$ is the span of certain vectors in $\mathbb{F}^m$, it is a subspace of $\mathbb{F}^m$.

Systems of linear equations (MATH1131 revision)

■ A matrix is in row echelon form (REF) if:

- all rows of zeros are at the bottom, and
- each leading entry is further to the right than all leading entries in the rows above it,

where the leading entry of a non-zero row is the leftmost non-zero entry in that row. A column containing a leading entry is called a leading column.

■ The nature of solutions of a system of linear equations $A\mathbf{x} = \mathbf{b}$ (for some $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{x} \in \mathbb{R}^n$, and $A \in M_{m,n}(\mathbb{R})$) can be found once its corresponding augmented matrix $(A \,|\, \mathbf{b})$ is reduced to row echelon form $(U \,|\, \mathbf{y})$.

- If the last column $\mathbf{y}$ is a leading column, then the system has no solutions.
- Otherwise, if the last column $\mathbf{y}$ is not a leading column, the system has at least one solution:
  - If all of the columns of U are leading columns, then the system has a unique solution.
  - If any of the columns of U are not leading columns, then the system has infinitely many solutions. If there are exactly k non-leading columns in U, then the solution set is determined by exactly k parameters.